

May 28, 2026

Dear Editors,

We respectfully submit our manuscript entitled “**CertiProof: A Hybrid Propositional Proof System with Adaptive Tactic Synthesis and Certified Proof Checking**” for consideration at **ACM Transactions on Computational Logic (TOCL)**.

What CERTIPROOF is (and is not). CERTIPROOF is a *certified hybrid proof system* that combines automated SAT solving with interactive theorem proving through adaptive tactic synthesis. We must emphasise at the outset: **CERTIPROOF is NOT a SAT solver**. It is a *proof generator* whose primary objectives are (i) producing independently verifiable, human-readable natural deduction proofs under the de Bruijn criterion, and (ii) enabling automated tactic learning without pre-trained data. Raw solving speed is a secondary concern that we measure honestly but do not optimise for. Confusing CERTIPROOF with a SAT solver would miss its entire contribution.

Unique value proposition. To the best of our knowledge, CERTIPROOF is **the only system that produces certified, human-readable natural deduction proofs with zero pre-training**. The system operates from the very first proof attempt—no offline corpus, no curated training data, no pre-computed tactic library. It learns *during* the proof-search session via EXP3-ATSS, an adversarial-bandit framework with provable $O(\sqrt{KT \ln K})$ regret guarantees under non-i.i.d. formula sequences. All neural and neural-symbolic alternatives (DeepSeek-Prover, AlphaProof, DreamProver, LeanAgent) require substantial offline pre-training on curated proof corpora; CERTIPROOF eliminates this dependence entirely.

Double formalisation — soundness *and* completeness. A distinguishing strength of CERTIPROOF is its **double-track verification** in Rocq 9.0 with **zero admitted proofs**:

1. **Soundness** (full mechanisation): every inference rule in the hybrid calculus (17 ND rules, sequent calculus bridge, 5 novel rules) is proven to preserve semantic validity via structural induction in Rocq, with a bridge lemma connecting Boolean and Prop-level semantics across all 7 connectives (1171 lines, 0 admits).
2. **Kálmár constructive completeness** (full mechanisation): every classical tautology admits a natural-deduction proof from the empty context, proved constructively via the signed-literal method with variable-elimination induction over excluded middle. This establishes completeness *internal* to the ND calculus, without reliance on external proof systems.

Together, these two theorems provide a *two-tier* completeness guarantee—the ND fragment alone is absolute-complete (Kálmár), while the full CERTIPROOF calculus is efficient-complete (p-simulation of Frege). We are not aware of any other proof system with both soundness and constructive completeness machine-checked in a proof assistant at this level of detail.

Experimental methodology: real wall-clock measurements. All performance data reported in this manuscript are obtained from *real wall-clock measurements* on standard hard-

ware (Intel Core i7-13700K, 32 GB DDR5 RAM, Windows 11, Python 3.12). Each experiment is repeated at least 5 times, and we report mean, standard deviation, minimum, and maximum values. We compare CERTIPROOF against Glucose4 on equivalent problem instances, explicitly acknowledging the Python/C++ implementation gap as an implementation artifact rather than an algorithmic limitation. This methodology is standard in the SAT community and ensures reproducibility.

Addressing the speed comparison with industrial solvers. We anticipate that reviewers will ask about the runtime gap between CERTIPROOF (Python 3.12) and industrial SAT solvers (C/C++). We address this head-on:

CERTIPROOF is NOT a SAT solver — it is a certified proof generator.

The $\sim 200\times$ Python/C++ gap is an *implementation artifact*, not an algorithmic deficiency. Factor out language overhead, and CERTIPROOF’s CDCL kernel performs within a factor of 25–35 of Kissat on equivalent operation counts. The remaining gap is the price of certification: every conflict produces a Craig interpolant and a verified proof step—operations with no counterpart in conventional solvers. In safety-critical domains where the question “is this verdict correct?” outweighs “how fast was it obtained?”, we believe a certified proof is categorically more valuable than a faster unverified answer.

Architectural innovation: the virtuous cycle. The central architectural insight of CERTIPROOF is the *virtuous cycle* that links its four components: (1) CDCL conflict analysis produces learned clauses, which feed (2) incremental Craig interpolation to extract semantically meaningful interpolants, which populate (3) the ATSS lemma table to guide tactic selection, whose successful proofs in turn (4) provide reward signals back to ATSS, closing the loop. This integrated feedback architecture distinguishes CERTIPROOF from systems that treat search, learning, and verification as separate modules. The cycle improves monotonically within a single proof-search session—no offline retraining needed.

Summary of contributions.

1. *A hybrid proof calculus* unifying natural deduction, sequent calculus, and resolution with five novel, soundness-verified inference rules.
2. *EXP3-ATSS*, the first application of adversarial bandit theory to proof search, with provable regret bounds.
3. *Mechanically verified soundness and Kálmár constructive completeness* in Rocq 9.0.
4. *Real-wall-clock experimental evaluation* across 8 benchmarks with standard hardware measurements, explicitly distinguishing implementation overhead from algorithmic performance.
5. *Comprehensive evaluation* covering classical tautologies, pigeonhole principles, random 3-CNF phase transitions, proof quality, ablation, scalability, SOTA comparison, verification-relevant benchmarks, and neural tactic selection.

Relevance to TOCL. CERTIPROOF makes contributions at the intersection of proof com-

plexity (the hybrid calculus and its p-simulation of Frege), automated reasoning (CDCL with incremental interpolation and EXP3-ATSS), and formal verification (full mechanisation of soundness and Kálmár completeness in Rocq). This combination—theoretical proof systems, practical certified proving, and machine-checked metatheory—aligns closely with TOCL’s scope, which emphasises computational logic in its broadest sense.

Revision notes. This manuscript has been revised in response to detailed peer feedback. Key improvements include: (i) replacement of synthetic operation-primitive estimates with real wall-clock measurements on standard hardware; (ii) expanded benchmarks including larger random 3-CNF instances and verification-relevant benchmarks; (iii) a substantially extended Related Work section covering neural/ML-guided theorem proving (TacticToe, DeepSeek-Prover), verified SAT solvers (Fleury et al. 2025, Lammich LRAT), and interactive ATP systems; (iv) clarified system positioning as a certified proof system (not a SAT solver) with ND proof readability demonstrations; and (v) discussion of extended resolution and future performance optimisation paths (Rust/C++ FFI, JIT compilation).

We believe CERTIPROOF offers a novel perspective at the intersection of proof complexity, automated reasoning, and formal verification, and we hope you will find it suitable for your venue.

Respectfully,

The Authors